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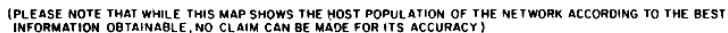
IPv4 ADDRESSING

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Protocol wars

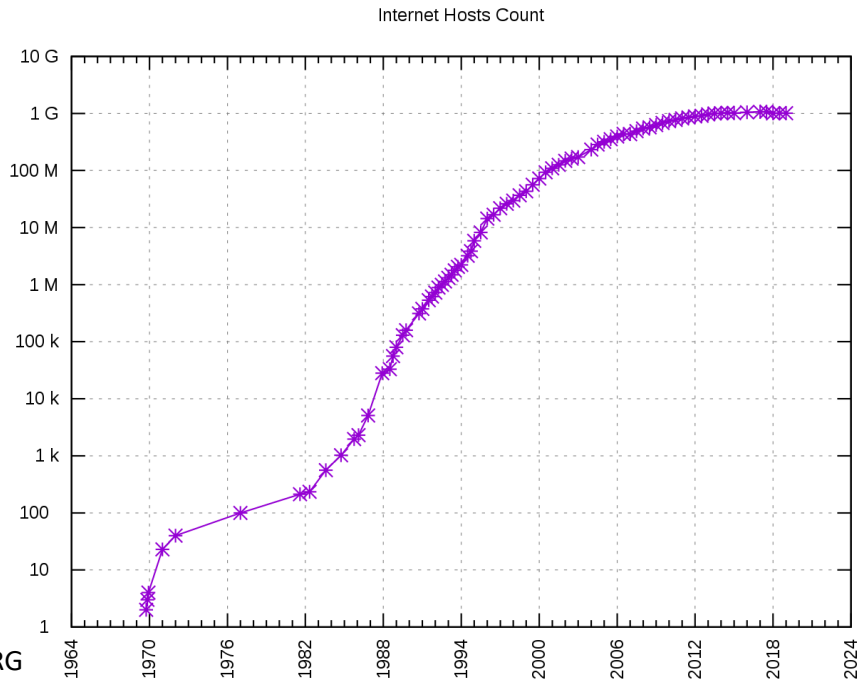
1977 >>

1977 >>



NAMES SHOWN ARE IMP NAMES, NOT (NECESSARILY) HOST NAMES

Development of internet network



Źródło: WIKIPEDIA.ORG

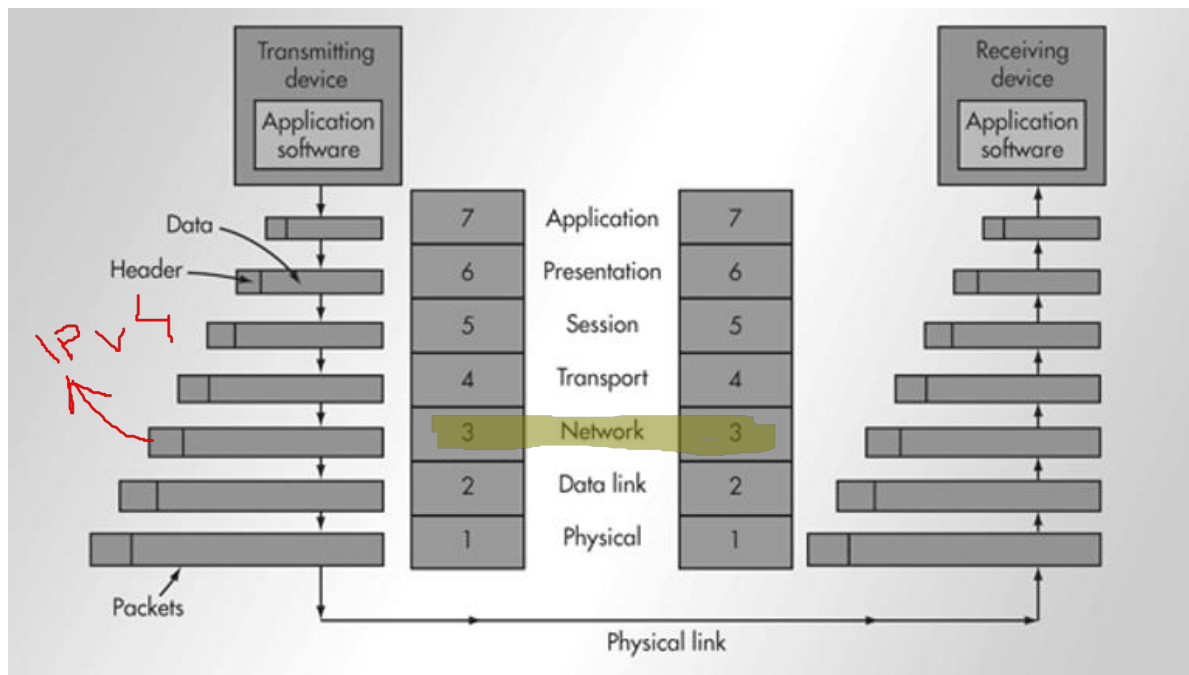
Protocol wars

- **1969** the ARPANET network was created - beginnings of networks
- **Before 1981** in IP addresses only the first 8 bits represented the network,
- **Since 1981** A 32-bit IP address allows the use of three classes (RFC 791),
- **Before 1983** – NCP (Network Control Protocol)
- **1 | 1983** – ARPANET changed its core networking protocols from NCP to the more flexible and powerful TCP/IP („a flag day”)
- **1993** classless inter-domain routing created CIDR (RFC 1519),

https://en.wikipedia.org/wiki/Protocol_Wars

[https://en.wikipedia.org/wiki/Flag_day_\(computing\)](https://en.wikipedia.org/wiki/Flag_day_(computing))

IPv4 addresses



Network layer

Network layer:

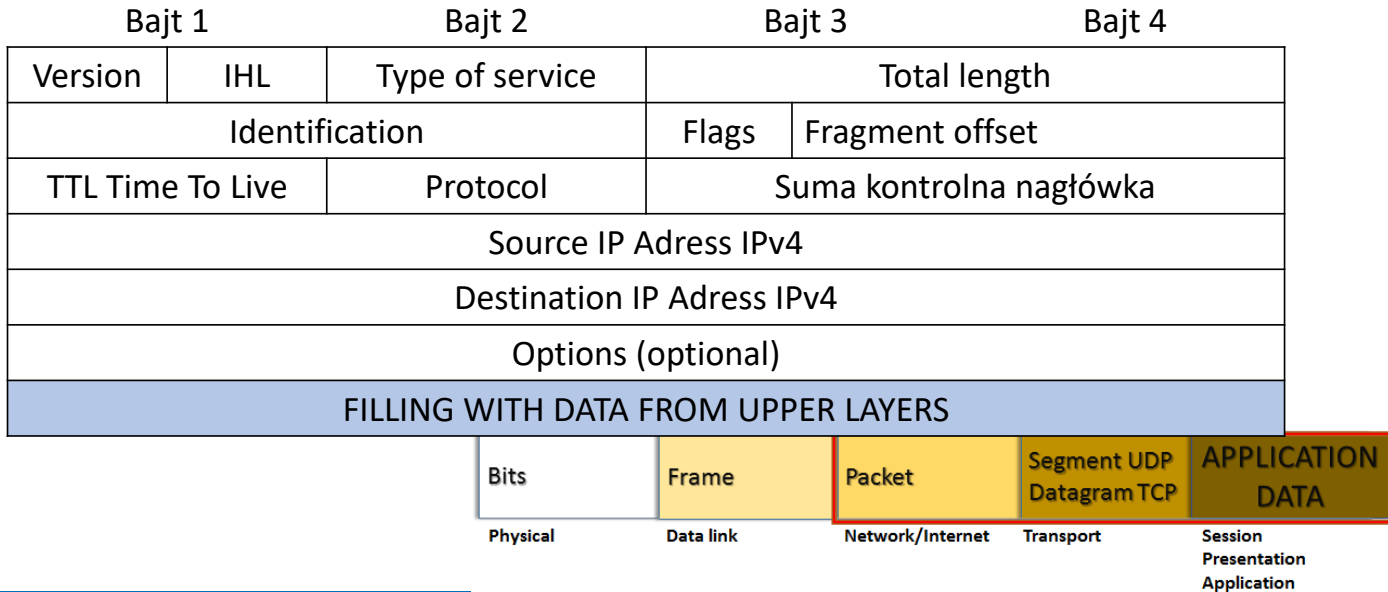
- Provides a connection between the sender and receiver hosts,
- Each sending and receiving device has a unique address (e.g. IPv4),

The network layer performs the key task of routing in communication between networks – finding the best route for a packet in the network.

Network protocols

	MODEL OSI	PDU	Addres	Device	Protocols	MODEL TCP
Implementation hardware	7 Application				HTTP, DNS, SMTP, POP, SSL, SQL, DHCP	Application
	6 Presentation					
	5 Session					
	4 Transport	Segment TCP Datagram UDP	Port Number		TCP, UDP, SCTP	Transport
	3 Network	Packet	IP address	Router (Ruter)	IPv4, IPv6, ICMP	Internet
	2 Link layer	Frame	Physical address MAC	Switch (Przełącznik)	ARP, Ethernet, PPP, CDP	Network Access
	1 Physical			Hub (Koncentrator)	Bluetooth, DSL, 802.11	

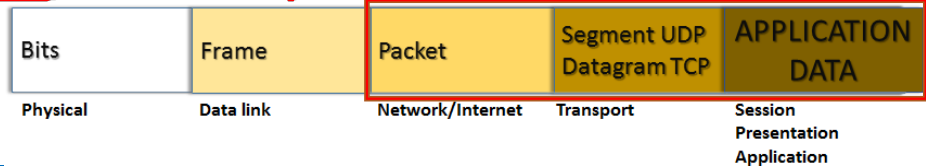
IPv4 header



Adres IPv4, Network mask

IP address 192.168.1.45			
192	168	1	45
11000000	10101000	00000001	00101101

Network mask 255.255.255.0 or ____/24			
255	255	255	0
11111111	11111111	11111111	00000000



Class addressing

class of
IP address
defines
network
mask

Class	First octet	IP address range	Net mask	Number of networks	Number of hosts
A	0 _ _ _	<u>0.0.0.0 - 127.255.255.255</u>	<u>255.0.0.0</u>	127	16.777.214
B	10 _ _	<u>128.0.0.0 - 191.255.255.255</u>	255.255.0.0	16.382	65.534
C	110 _	<u>192.0.0.0 - 223.255.255.255</u>	<u>255.255.255.0</u>	2.097.150	254
D	1110	<u>224.0.0.0 - 239.255.255.255</u>	for multicast transmission		
E	1111	<u>240.0.0.0 - 255.255.255.255</u>	Reserved for IETF		

Adresy localhost (loopback)

IP: 127.0.0.1

<https://en.wikipedia.org/wiki/Localhost>

```
H:\>ping localhost
```

```
Pinging PK [::1] with 32 bytes of data:
```

```
Reply from ::1: time<1ms
```

```
Reply from ::1: time<1ms
```

```
Reply from ::1: time<1ms
```

```
Reply from ::1: time<1ms
```

```
Ping statistics for ::1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
H:\>ping 127.0.0.1
```

```
Pinging 127.0.0.1 with 32 bytes of data:
```

```
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

```
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

```
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

```
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 127.0.0.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
H:\>
```

Address 0.0.0.0

0.0.0.0

- Undefined address IP

0.0.0.0/0

- IP block containing all IP addresses
- This is how a static route is marked

<https://en.wikipedia.org/wiki/0.0.0.0>

IPv4 link-local

automatic private IP addressing (APIPA)

- 169.254.0.0/16 -> 169.254.0.0 – 169.254.255.255

Local addresses



class	Range of IP addresses	Network mask	CIDR notation
A	10.0.0.0 – 10.255.255.255	255.0.0.0	10.0.0.0/8
B	172.16.0.0 – 172.31.255.255	255.240.0.0	172.16.0.0/12
C	192.168.0.0 – 192.168.255.255	255.255.0.0	192.168.0.0/16

<https://whatismyipaddress.com/private-ip>

IPv4 multicast

Selected IP addresses of the class D (range 224.0.0.0 - 239.255.255.255)

- 224.0.0.0/24 only used to connect to a local multicast router
- 232.0.0.0/8 is reserved for multicast from the selected source (SSM).
- 233.0.0.0/8 is disconnected from addresses GLOP

Reservation of all Class D IP addresses:

<https://www.iana.org/assignments/multicast-addresses/multicast-addresses.xhtml>

Adres IPv4 – types of broadcast

Directed broadcast IPv4 address

- Has a specific network part, and in the host part it has all ones,
- If the network address is known - 172.16.255.255/16
- Used in ARP and DHCP.

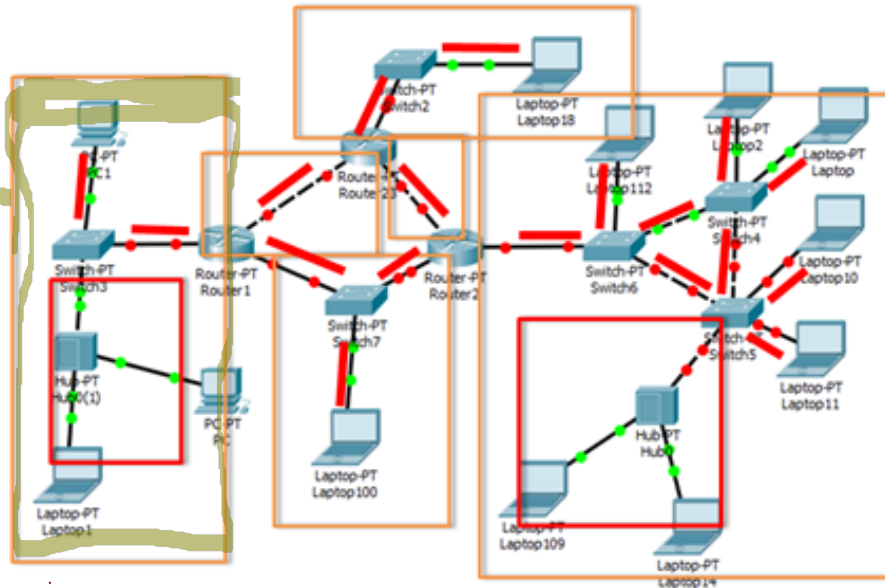
Limited broadcast IPv4 address

- An address where all 32 bits are set to one – 255.255.255.255,
- In case the network address is unknown,

Broadcast domains vs collision domains

Broadcast domains
(domeny rozgłoszeniowe)

Collision domains
(domeny kolizji)



IPv4 ADDRESSING – tasks

Converting a mask to slash notation

What is the slash form of a decimal mask?

• $\frac{255}{8}.\frac{255}{8}.0.0 \rightarrow /16$

• $\frac{255}{8}.\frac{255}{8}.\frac{192}{2}.0 \rightarrow /18$

• $\frac{255}{8}.\frac{255}{8}.\frac{255}{8}.\frac{240}{4} \rightarrow /28$

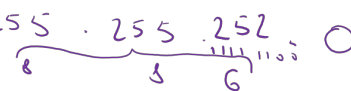
• $\frac{255}{3}.\frac{255}{3}.\frac{255}{3}.\frac{252}{4} \rightarrow /30$

• $\frac{255}{252}.\frac{255}{11111111}.\frac{254}{11111100} \rightarrow /31$

Przeliczanie maski do postaci dziesiętnej

What is the decimal form of a slash mask?

- $______ / 24 \rightarrow 255.255.255.0$

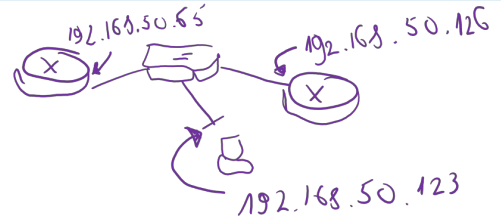
- $______ / 22 \rightarrow 255.255.252.0$

- $______ / 23 \rightarrow 255.255.254.0$
- $______ / 25 \rightarrow 255.255.255.128$


Identification I

IP address: **192.168.50.123**

Net mask: **255.255.255.192** or /

A	B	C
0	0	0
1	1	0
1	0	1
0	0	0



IP address:	1100 0000 . 1010 1000 . 0011 0010 . 01	11 1011
Net mask :	1111 1111 . 1111 1111 . 1111 1111 . 11	00 0000
Network addr.:	1100 0000 . 1010 1000 . 0011 0010 . 01	00 0000
First host addr.:	1100 0000 . 1010 1000 . 0011 0010 . 01	00 0001
..		
Last host addr:	1100 0000 . 1010 1000 . 0011 0010 . 01	11 1110
broadcast addr:	1100 0000 . 1010 1000 . 0011 0010 . 01	11 1111

Provide:

- Network address: **192.168.50.64**
- First host address: **192.168.50.65**
- The last host address: **192.168.50.126**
- Broadcast address: **192.168.50.127**
- The number of hosts: **$2^{(zeros_in_the_net_mask)} - 2 = 2^6 - 2 = 64 - 2 = 62$**

Identification II

IP address **172.16.139.5** Net mask: **/18 = 255.255.192.0**

IP address:	10101100.00010000.10		001011.00000101
Net mask:	11111111.11111111.11		000000.00000000
Net addr:	10101100.00010000.10		000000.00000000
First host:	10101100.00010000.10		000000.00000001
hosts...	10101100.00010000.10		101010.11101101
Last host:	10101100.00010000.10		111111.11111110
Broadcast:	10101100.00010000.10		111111.11111111

- Network address: **172.16.128.0**
- First host address: **172.16.128.1**
- The last host address: **172.16.191.254**
- Broadcast address: **172.16.191.255**
- The number of hosts $2^{(\text{zeros in the net mask})} - 2 = 2^{14} - 2 = 16382$

CIDR, VLSM

Class addressing

*this standard
used before CIDR*

Class	First octet	IP address range	Net mask	Number of networks	Number of hosts
A	0 _ _ _	<u>0</u> .0.0.0 - <u>127</u> .255.255.255	<u>255.0.0.0</u>	127	16.777.214
B	<u>10</u> _ _	<u>128</u> .0.0.0 - <u>191</u> .255.255.255	<u>255.255.0.0</u>	16.382	65.534
C	<u>1110</u> _	<u>192</u> .0.0.0 - <u>223</u> .255.255.255	<u>255.255.255.0</u>	2.097.150	254
D	1 1 1 0	<u>224</u> .0.0.0 - <u>239</u> .255.255.255	for multicast transmission		
E	1 1 1 1	<u>240</u> .0.0.0 - <u>255</u> .255.255.255	Reserved for IETF		

CIDR (Classless Inter-Domain Routing)

CIDR (Classless Inter-Domain Routing)

- Classless IP routing method between domains
- introduces a variable mask length (VLSM), which enables network aggregation and division of the network into subnets,
- Introduced in 1993 by the IETF (Internet Engineering Task Force) to replace the earlier, classful addressing method,

Number of subnetworks for network in class B

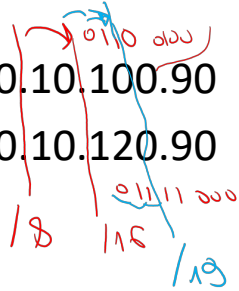
Network address: 172.16.0.0 | Network mask: 255.255.0.0 $\Leftrightarrow$ __.__.__.__/16

Network address	Network mask		Number of subnets	Number of host addresses
172.16.0.0	255.255.192.0	/18 .11 000000.00000000	2	16.382
172.16.0.0	255.255.224.0	/19 .111 00000.00000000	6	8.190
172.16.0.0	255.255.240.0	/20 .1111 0000.00000000	14	4.094
172.16.0.0	255.255.248.0	/21 .11111 000.00000000	30	2.046
172.16.0.0	255.255.252.0	/22 .111111 00.00000000	62	1.022
172.16.0.0	255.255.254.0	/23 .1111111 0.00000000	126	510
172.16.0.0	255.255.255.0	/24 .11111111. 00000000	254	254
172.16.0.0	255.255.255.128	/25 .11111111.1 00000000	510	126
172.16.0.0	255.255.255.192	/26 .11111111.11 00000000	1.022	62
172.16.0.0	255.255.255.224	/27 .11111111.111 00000000	2.046	30
172.16.0.0	255.255.255.240	/28 .11111111.1111 00000000	4.094	14
172.16.0.0	255.255.255.248	/29 .11111111.11111 00000000	8.190	6
172.16.0.0	255.255.255.252	/30 .11111111.111111 00000000	16.382	2

VLSM - tasks

Suggest a suitable network for hosts I

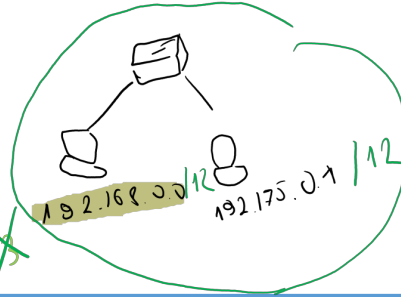
- 10.10.100.90
- 10.10.120.90



decimal	Binary	
10.10.100.90	? 10.10.0110 0100, 90	First host address
10.10.120.90	? 10.10.0111 1000, 90	Second host address
?	? 255.255.224.0	Net mask
?	? 10.10.0110 1000, 96	Network address

Suggest a suitable network for hosts III

- 192.168.0.0
- 192.175.0.1



NA:
192.160.0.0/12

decimal	Binary	
192.168.0.0	192. 168 . 0 . 0	First host address
192.175.0.1	192. 175 . 0 . 1	Second host address
		Net mask
	192. 168 . 0 . 0	Network address
		First host address

VLSM dynamic subnets – example I

We have a network:

- Net address **192.168.1.0**
- Mask **255.255.255.0 /24**

Division into subnets :

- A – 100 hosts,
- B – 50 hosts,
- C – 50 hosts,

- Network address: **192.168.1.0**
- First host address: **192.168.1.1**
- Last host address: **192.168.1.254**
- Broadcast address: **192.168.1.255**
- Number of hosts: $2^{(\text{nr_of_0_in_network_mask})} - 2 = 256 - 2 = 254$

$$\begin{array}{r} 192.168.1.0 \\ 255.255.255.0 \\ \hline N, 192.168.1.0 \\ 192.168.1.1 \\ 192.168.1.254 \\ 192.168.1.255 \end{array}$$

$$\begin{array}{l} 2^7 - 2 = 126 \quad \checkmark \\ \quad \quad 128 \quad \quad 2^6 - 2 \\ A \Rightarrow 0 \quad \quad 128 \downarrow \\ \quad \quad 127 \end{array}$$

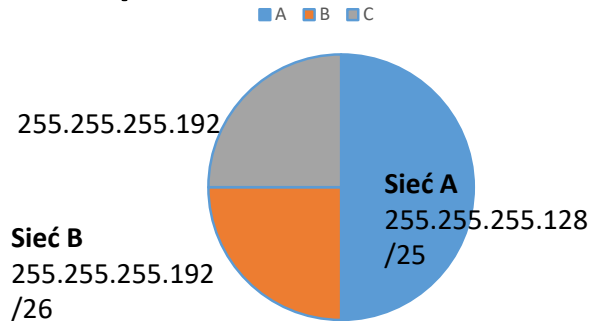
VLSM dynamic subnets – example I

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- Net address **192.168.1.0**
- Mask **255.255.255.0 /24**

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	Mask - decimal	Net address	Broadcast address	Nets/hosts
A				
B				
C				

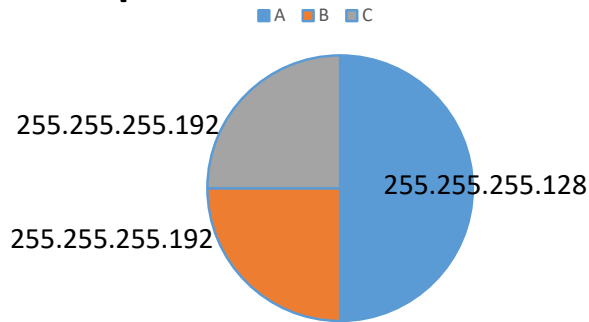
VLSM dynamic subnets – example I

We have a network:

- IP address **192.168.1.0**
- Mask **255.255.255.0 /24**

Division into subnets :

- A – 100 hosts,
- B – 50 hosts,
- C – 50 hosts,



NETWORK C (50 hosts)

192.168.1.192/26 – network address
192.168.1.255 – broadcast

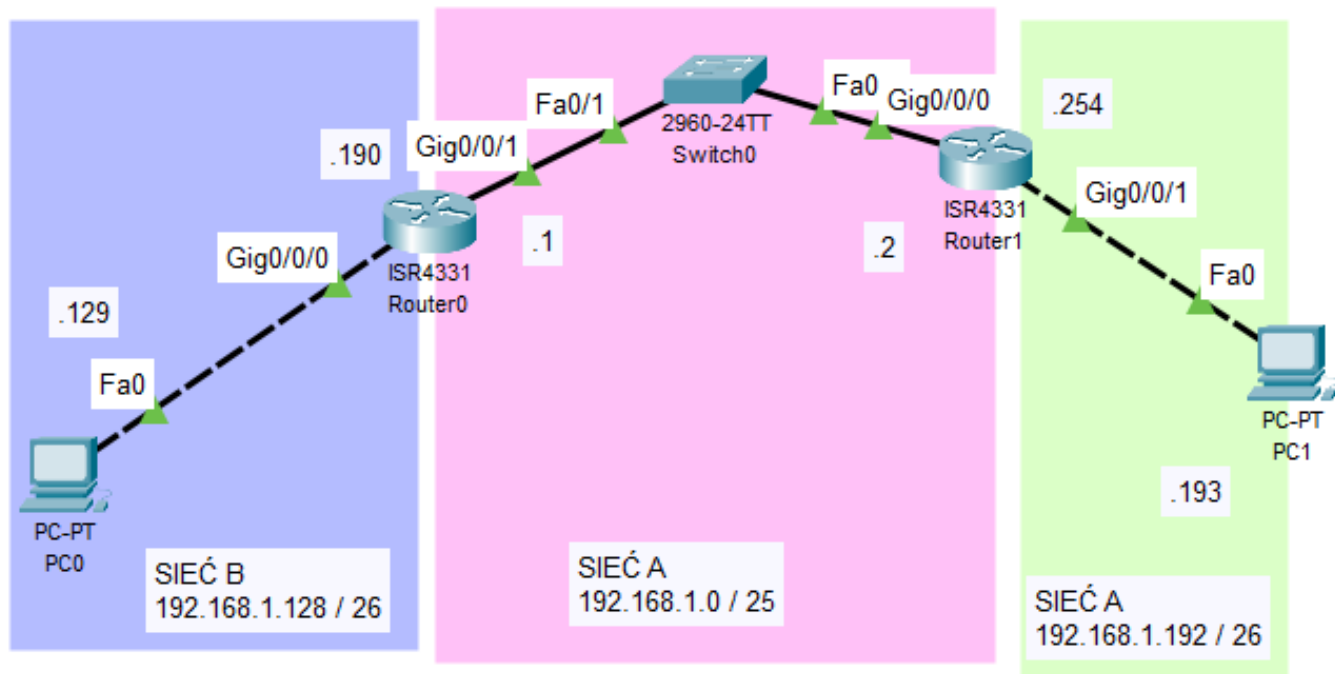
NETWORK B (50 host)

192.168.128.0/26 – network address
192.168.1.191 – broadcast

NETWORK A (100 hosts)

192.168.1.0/25 – network address
192.168.1.127 – broadcast

VLSM dynamic subnets – example I

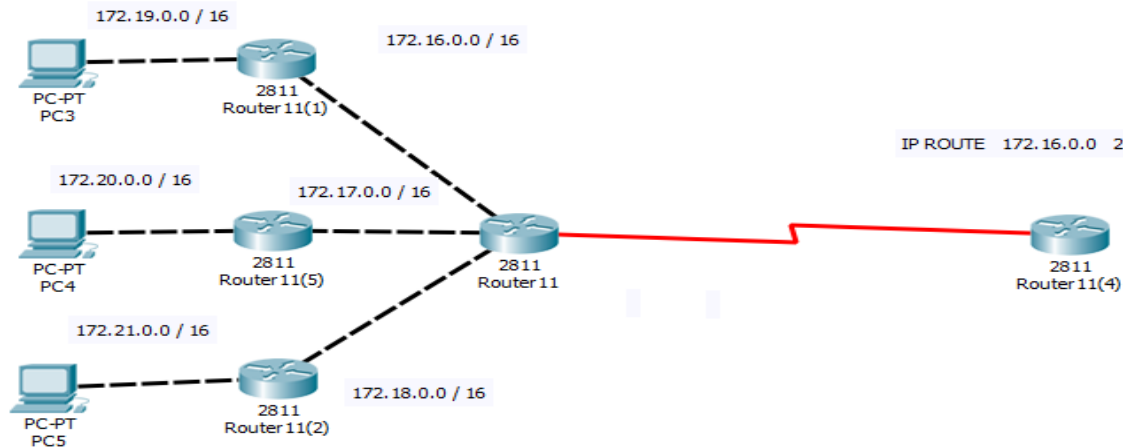


Network summary

Network summary

Propose a network that aggregates the following networks:

- 172.16.0.0 /16
- 172.17.0.0 /16
- 172.18.0.0 /16
- 172.19.0.0 /16
- 172.20.0.0 /16
- 172.21.0.0 /16



Network summary

IP address	IP address - binary	
172.16.0.0 /16	10101100	00010000
172.17.0.0 /16		00010001
172.18.0.0 /16		00010010
172.19.0.0 /16		00010011
172.20.0.0 /16		00010100
172.21.0.0 /16		00010101
	172	0 0

Summarized network:

172.16.0.0 /13

New network map:

255.248.0.0

